Aref Pariz, Ph.D.

Assistant Instructor, Department of Psychiatry, UT Southwestern Medical Center
Dallas, TX, USA

PROFESSIONAL PROFILE

Computational, theoretical, and translational neuroscientist with training in Physics and postdoctoral experience in brain stimulation, neural dynamics, TMS-EEG, tACS, effective connectivity, and individualized modeling of brain networks. Current research focuses on computational psychiatry and precision non-invasive brain stimulation in clinical populations.

RESEARCH AREAS

Computational psychiatry and computational neuroscience

Non-invasive brain stimulation: TMS, theta burst stimulation, tACS, and tES/tCS

Neural dynamics, oscillations, information transfer, and effective connectivity

TMS-EEG, EEG analysis, fMRI-constrained modeling, and network neuroscience

Synaptic plasticity, neural heterogeneity, resilience, and brain-state dependence

Machine learning, data analysis, and individualized stimulation protocols

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Feb. 2026-present	**Assistant Instructor, Department of Psychiatry, UT Southwestern Medical Center, Dallas, TX, USA** Research focus: individualized tES/tACS and related stimulation methods for addictive and related conditions.
Sep. 2023-Jan. 2026	**Postdoctoral Researcher, Institute of Mental Health Research at The Royal and University of Ottawa, Ottawa, ON, Canada** Mentors: Dr. Sara Tremblay and Prof. Jeremie Lefebvre. Research focus: effects of TMS and theta burst stimulation on brain disorders; TMS-EEG analysis and effective connectivity.
Jan. 2021-Aug. 2023	**Postdoctoral Researcher, Department of Biology, University of Ottawa, Ottawa, ON, Canada** Mentor: Prof. Jeremie Lefebvre. Research focus: effects of brain stimulation on synaptic modifications and neuronal population dynamics.
Dec. 2019-Dec. 2020	**Postdoctoral Researcher, Department of Chemistry, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran** Mentor: Prof. Abdollahi. Research focus: chemometric data analysis methods in neuroscience.
Apr. 2018-Mar. 2019	**Visiting Researcher, IFISC (UIB-CSIC), Palma de Mallorca, Spain** Mentor: Prof. Claudio R. Mirasso. Research focus: effects of communication delay on information transfer.
Sep. 2012-Mar. 2018	**Ph.D. Researcher, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran** Mentor: Prof. Alireza Valizadeh.
Sep. 2009-Mar. 2012	**M.Sc. Researcher, University of Zanjan, Zanjan, Iran** Mentor: Dr. Saeed Ghanbari.

EDUCATION

2012-2018	**Ph.D. in Physics, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran** Dissertation: Controlling the information transfer by phase manipulation of population activity of neuronal networks. Supervisor: Dr. Alireza Valizadeh.
2009-2011	**M.Sc. in Physics, University of Zanjan, Zanjan, Iran** Thesis: Collapse of Dipolar Bose-Einstein Condensation. Supervisor: Dr. Saeed Ghanbari.
2005-2009	**B.Sc. in Physics, Shahid Bahonar University of Kerman, Kerman, Iran**

FUNDING AND FELLOWSHIPS

2026	**NIH R01** Title: Optimized Frontoparietal Synchronization to Reduce Craving in Methamphetamine Use Disorder: A Concurrent tACS-fMRI Trial. Role: Key Personnel. Status: Under Review.
2019	**Postdoctoral Fellowship Grant, Iranian Elite Federation**
2018	**Visiting Fellowship Grant, IFISC, University of the Balearic Islands, Palma de Mallorca, Spain**
2017	**Study Grant, Institute for Research in Fundamental Sciences, Tehran, Iran**

CLINICAL STUDIES AND RESEARCH PROTOCOLS

tACS-fMRI (NEC-TACS)
UT Southwestern Medical Center, Dallas, TX, USA
Role: Co-Investigator and Study Coordinator.
Status: IRB approved, August 2026. Preparing to recruit.

Frontopolar tDCS for Drug Craving
UT Southwestern Medical Center, Dallas, TX, USA
Role: Co-Investigator and Study Coordinator.
Status: Submitted for IRB review, September 2026.

Dallas-1000
UT Southwestern Medical Center, Dallas, TX, USA
Role: Research Assistant.
Status: Pre-submission.

Closed-loop FDCR-tACS
UT Southwestern Medical Center, Dallas, TX, USA
Role: Co-Investigator and Study Coordinator.
Status: Under preperation.

PEER REVIEWED PUBLICATIONS AND ACCEPTED MANUSCRIPTS

- O'Sullivan, D., **Pariz, A.**, Trotter, D., Al Dahhan, N.Z., Knowles, J.K., Frankland, P.W., Mabbott, D. and Lefebvre, J., (2026). A Computational Framework for Myelin as a Regulator of Temporal Integration, Neuronal Excitability, and Synaptic Plasticity. *Glia*, *74*(12), p.e70231. 10.1002/glia.70231
- Hutt, A., McCulloch, M., Hudetz, A.G., **Pariz, A.**, and Lefebvre, J., (2026). Noise-like fluctuations drive shifts in neuronal timescales and 1/f power spectra across brain states. *PLOS Complex Systems*, *3*(9), p.e0000133. https://doi.org/10.1371/journal.pcsy.0000133
- Sharifzadeh, S., Jouzdani, A. F., Souki, A., Brunoni, A. R., Marques, D. B., Ebrahimi, M., Ghobadi-Azbari, P., **Pariz, A.**, Soleimani, G., Yavari, F., Khosrowabadi, R., Violante, I. R., & Ekhtiari, H. (2026). Personalized Brain Stimulation based on Brain Oscillations: A Systematic Review of MEG- and EEG-informed Transcranial Alternating Current Stimulation (tACS). *Neuroscience & Biobehavioral Reviews*, 106991. https://doi.org/10.1016/j.neubiorev.2026.106991
- Asl, K. J., Naseri, A., Amjadi, M., Alavi, S. M. S., **Pariz, A.**, Tatardar, F., & Rasoulzadeh, F. (2026). Nanozymatic platform for dual colorimetric and fluorimetric sensing of H_2O_2 and glucose using Ni-$CoWO_4$ nanoparticles. *Scientific Reports*. https://doi.org/10.1038/s41598-026-58001-3
- Lefebvre, J., **Pariz, A.**, Thivierge, J.-P., & Hutt, A. (2025). Surfing brain waves without sinking: A commentary on "Entrainment by transcranial alternating current stimulation: Insight from models of cortical oscillations and dynamical systems theory" by M. Madadi Asl and A. Valizadeh. Physics of Life Reviews, 55, 55-57. https://doi.org/10.1016/j.plrev.2025.08.009
- Trotter, D. J., **Pariz, A.**, Hutt, A., & Lefebvre, J. (2025). Morphological variability may limit single-cell specificity to electric field stimulation. Frontiers in Synaptic Neuroscience, 17, 1621352. https://doi.org/10.3389/fnsyn.2025.1621352
- Lefebvre, J., & **Pariz, A.** (2025). Amplifying post-stimulation oscillatory dynamics by engaging synaptic plasticity with transcranial alternating current stimulation. Frontiers in Network Physiology, 5, 1621283. https://doi.org/10.3389/fnetp.2025.1621283

- Hutt, A., Trotter, D., **Pariz, A.**, Valiante, T. A., & Lefebvre, J. (2024). Diversity-induced trivialization and resilience of neural dynamics. Chaos: An Interdisciplinary Journal of Nonlinear Science, 34(1), 013147. https://doi.org/10.1063/5.0165773
- **Pariz, A.**, Trotter, D., Hutt, A., & Lefebvre, J. (2023). Selective control of synaptic plasticity in heterogeneous networks through transcranial alternating current stimulation (tACS). PLOS Computational Biology, 19(4), e1010736. https://doi.org/10.1371/journal.pcbi.1010736
- Yousefi, N. N., Mortezapour, A., Naeimi, G., Nosrati, F., **Pariz, A.**, & Lo Franco, R. (2022). Quantum enhancement of qutrit dynamics through driving field and photonic-band-gap crystal. Physical Review A, 105(4), 042212. https://doi.org/10.1103/PhysRevA.105.042212
- Sanchez-Claros, J., **Pariz, A.**, Valizadeh, A., Canals, S., & Mirasso, C. R. (2021). Information transmission in delay-coupled neuronal circuits in the presence of a relay population. Frontiers in Systems Neuroscience, 15, 705371. https://doi.org/10.3389/fnsys.2021.705371
- **Pariz, A.**, Fischer, I., Valizadeh, A., & Mirasso, C. R. (2021). Transmission delays and frequency detuning can regulate information flow between brain regions. PLOS Computational Biology, 17(4), e1008129. https://doi.org/10.1371/journal.pcbi.1008129
- **Pariz, A.**, Esfahani, Z. G., Parsi, S. S., Valizadeh, A., Canals, S., & Mirasso, C. R. (2018). High frequency neurons determine effective connectivity in neuronal networks. NeuroImage, 166, 349-359. https://doi.org/10.1016/j.neuroimage.2017.11.014
- Mohammadi, A., Ghanbari, S., & **Pariz, A.** (2013). A two-dimensional permanent magnetic lattice for ultracold atoms. Physica Scripta, 88(1), 015601
- **Pariz, A.** Naiemi, G., (2011). The Student Guide to General Physics. Book in Persian

SUBMITTED MANUSCRIPTS, PREPRINTS, AND MANUSCRIPTS IN PREPARATION

- **Pariz, A.**, Hadas, H., Lepage, M., Daskalakis, Z. J., Lefebvre, J., & Tremblay, S.. Network-Level Responses to iTBS Pulse Dose: A Non-Linear Pattern Detected with TMS-EEG. *Under review*
- Hashemi, M., Shi, W., Soleimani, G., Villanueva-Miranda, I., **Pariz, A.,** Anteraper, S., Jin, J., Mansour L., S., Benette Marques, D., Mirjalili, M., Opitz, A., Juliano, A., Triebkorn, P., Wang, H., Yang, C., Rajji, T., Brunoni, A. R., Jirsa, V., Xiao, G., & Ekhtiari, H.. Artificial Intelligence for Precision Brain Stimulation: From Neuroimaging to Foundation Models and Virtual Brains. Under review. PsyArXiv preprint. https://doi.org/10.31234/osf.io/qnjd6_v1
- **Pariz, A.,** Lefebvre, J., & Tremblay, S.. NESTApp: A GUI-Based MATLAB Application for Simplified TMS-EEG Preprocessing with EEGLAB Under review. SSRN preprint. https://ssrn.com/abstract=6204470
- **Pariz, A.**, Jin, J., Rashedi, A., and Ekhtiari, H., A Practical Framework for High-Definition tES Montage and Dose Optimization: From Anatomical ariability to Cohort-Level Design, with an Exemplar Application in Methamphetamine Use Disorder. *In preparation.*
- Jin, J., **Pariz, A.**, Rashedi, A., and Ekhtiari, H., Comparative Methodologies in fMRI-Informed TMS for Depression: A Systematic Review. *In preparation*
- A., Rashedi, Jin, J., **Pariz, A.**, and Ekhtiari, H., Just-in-Time Neuromodulation: A Precision, State-Dependent Paradigm for On-Demand Brain Intervention. *In Preparation.*
- **Pariz, A.**, Hadas, H., Lepage, M., Daskalakis, Z. J., Lefebvre, J., & Tremblay, S., Network-Level Responses to iTBS Pulse Dose: A Non-Linear Pattern Detected with TMS-EEG. *In review.*
- **Pariz, A.**, Bolhasani, E., Lefebvre, J., Valizadeh, A. Delay-dependent communication bias in collective phase response curves of hybrid excitatory–inhibitory populations. *In Preparation.*
- **Pariz, A.**, Jin, J., Rashedi, A., Ekhtiari, H., Methodological Reporting and Parameter Heterogeneity in Concurrent tES–fMRI Studies: A Systematic ContES-Based Review (2019–2026). *In Preparation.*

INNOVATION, RESEARCH AND EDUCATIONAL TOOLS

NESTApp — TMS–EEG Preprocessing Application
Developed a MATLAB application with a graphical interface to simplify TMS–EEG preprocessing using EEGLAB. The application brings preprocessing steps into an accessible workflow for researchers. https://parsecneuro.github.io/tools/nestapp/index.html

EEG Cap Viewer — Interactive Electrode Visualization
Developed an interactive 3D tool for visualizing EEG electrode locations, comparing cap layouts, and examining distances between electrodes. Includes information about approximate underlying cortical regions to support research and teaching. https://parsecneuro.github.io/tools/eeg-cap-viewer/index.html

Paper Review — PDF Reading and Annotation Tool
Developed a browser-based tool that displays two independently scrolling views of the same paper, allowing readers to examine the text alongside figures or other sections. Supports annotations, comments, and export of review notes. https://parsecneuro.github.io/tools/paper-review/index.html

CONFERENCE PROCEEDINGS AND PUBLISHED ABSTRACTS

6. Pariz, A., Lefebvre, J., & Valizadeh, A. (2021). Phase response curve of the excitatory-inhibitory neuronal populations and its role in the coherent oscillation of interconnected brain regions. Journal of Computational Neuroscience, 49(Suppl. 1), S188-S188

5. Sanchez-Claros, J., Mirasso, C. R., Kang, M., **Pariz, A.**, & Fischer, I. (2019). Information transmission in delay-coupled neuronal circuits. CNS2019, BMC Neuroscience, 20, P81

4. Mirasso, C. R., **Pariz, A.**, Canals, S., & Valizadeh, A. (2019). High frequency neurons help routing information in brain networks. CNS2019, BMC Neuroscience, 20, P320

3. Pariz, A., Parsi, S. S., & Valizadeh, A. (2016). High frequency neuron can facilitate propagation of signal in neural networks. CNS2016, BMC Neuroscience, 17, P42

2. Mohammadi, A., Ghanbari, S., & **Pariz, A.** (2010). Direct forced evaporative cooling in permanent magnetic microtraps. arXiv:1007.2912. https://arxiv.org/abs/1007.2912

CONFERENCE PRESENTATIONS, POSTERS, SCHOOLS, AND WORKSHOPS

- **SUBMITTED CONFERENCE ABSTRACTS**
  - Dunne, W., **Pariz, A.**, Jouzizadeh, M., Cummings, E., Letourneau, R., Bangoura, B.-K., Elsharif, A., Shim, S., Almatrafi, A., Fus, D., Brender, R., Antochi, R., McMurray, L., Chaves, A. R., & Tremblay, S. (2026). Day-one Circuit Reactivity Tracks Recovery Rate in Theta-Burst Stimulation for Treatment-Resistant Depression. Abstract submitted to the 7th International Brain Stimulation Conference, Lyon, France, February 21–24, 2027. Decision pending.
  - Cummings, E., **Pariz, A.**, Letourneau, R., Jouzizadeh, M., Bangoura, B.-K., Elsharif, A., Shim, S., Couture, S., Dunne, W., Almatrafi, A., Fus, D., Brender, R., Antochi, R., McMurray, L., Chaves, A. R., & Tremblay, S. (2026). Sex Differences in the Relationship Between Electric Field Modelling and Clinical Response to TBS. Abstract submitted to the 7th International Brain Stimulation Conference, Lyon, France, February 21–24, 2027. Decision pending.
  - **Pariz, A.**, Jin, J., Rashedi, A., & Ekhtiari, H. (2026). Simulation-guided selection of a dual-site HD-tACS montage for methamphetamine use disorder. Abstract submitted to the 7th International Brain Stimulation Conference, Lyon, France, February 21–24, 2027. Decision pending.

POSTERS

July 2026	**Poster, NYC Neuromodulation Conference**, NY, USA. **Pariz, A., Jin J., Rasgedi A., & Ekhtiari, H.,** Simulation-guided selection of a practical F3–CP5 high-definition tACS montage for frontoparietal stimulation in methamphetamine use disorder.
July 2026	**Poster, NYC Neuromodulation Conference, NY, USA.** Jin, J., **Pariz, A.**, & Ekhtiari, H., Comparative Methodologies in fMRI-Informed TMS for Depression: A Systematic Review. **Poster,** Rashedi, A.,

July 2026	**Poster**, **NYC Neuromodulation Conference, NY, USA. Pariz, A.**, Jin, J., & Ekhtiari, H., Methodological Reporting and Parameter Heterogeneity in Concurrent tES–fMRI Studies: A Systematic ContES-Based Review (2019–2026).
Sep. 2025	**Poster, BBI2025 for Non-Invasive Brain Stimulation and Imaging, London, UK** Using SimNIBS Electric-Field Variables to Predict Response to Theta Burst Stimulation Treatment in Males and Females with Treatment-Resistant Depression.
June 2025	**Poster, NeuMe Conference, The Royal Ottawa Mental Health Centre, Ottawa, ON, Canada** Theta Burst Stimulation Produces Comparable Clinical Response in Younger and Older Adults with Treatment-Resistant Depression Despite Age-Related Differences in Neuroplasticity.
Feb. 2025	**Poster, 6th International Brain Stimulation Conference, Kobe, Japan** Comparing Unilateral and Bilateral Sequential TBS on prefrontal activity: a TMS-EEG study.
May 2023	**Poster, 16th Canadian Neuroscience Meeting, Montreal, QC, Canada** Stabilizing post-stimulation oscillatory dynamics by engaging synaptic plasticity with tACS.
May 2022	**Poster, 15th Canadian Neuroscience Meeting, Toronto, ON, Canada** Selective engagement of synaptic coupling using tACS.
July 2021	**Poster, 30th Annual Computational Neuroscience Meeting, Online** Phase response curve of the excitatory-inhibitory neuronal populations and its role in the coherent oscillation of interconnected brain regions.
Dec. 2021	**Poster, Neuromatch Conference v4, Online** Selective engagement of synaptic coupling using tACS.
Oct. 2019	**Participant, IBRO-VLTP Courses in Neuroscience, Tarbiat Modares University, Tehran, Iran**
June 2018	**Poster, Third Workshop on Advanced Methods in Theoretical Neuroscience, Gottingen, Germany** High frequency neurons determine effective connectivity in neuronal networks.
Jan. 2018	**Poster, Computational Neuroscience Conference, IPM, Tehran, Iran** Manipulating phase relation between neuronal populations by time delay.
July 2016	**Poster, 25th Annual Computational Neuroscience Meeting, Jeju, South Korea** High frequency neuron can facilitate propagation of signal in neural networks.
May 2016	**Participant, 3rd Workshop in Computational Neuroscience, Babol, Iran**
Dec. 2015	**Participant, 4th Basic and Clinical Neuroscience Congress, Tehran, Iran**
Aug. 2014	**Participant, 4th Computational Neuroscience Workshop on Spikes, Synchronization and Cognition, IASBS, Zanjan, Iran**
Oct. 2013	**Participant, 3rd Tehran IBRO School of Neuroscience: Molecular, Electrophysiological and Behavioral Approaches, Tehran, Iran**
Dec. 2012	**Participant, 5th Conference in Statistical Physics, Soft Matter and Complex Systems, University of Tehran, Tehran, Iran**
Mar. 2012	**Poster, Young Atom Opticians Conference, Krakow, Poland** Single Permanent Magnetic Microtrap.
July 2010	**Poster, 22nd International Conference on Atomic Physics, Cairns, Australia** Single Permanent Magnetic Microtraps for Direct Forced Evaporative Cooling.
Jan. 2009	**Participant, Workshop on Quantum Computing and Quantum Information Processing, Shahid Beheshti University, Tehran, Iran**
Jan. 2009	**Participant, Symposium on Experimental Aspects of Quantum Computing, Shahid Beheshti University, Tehran, Iran**

INVITED TALKS AND ORAL PRESENTATIONS

June 2025	**Invited Speaker, University of Minnesota, Minneapolis, MN, USA** Talk: Modulating brain circuits and neuronal population activity through tACS.
Oct. 2023	**Speaker, Canadian Computational Neuroscience Spotlight, Canada** Talk: Selective control of synaptic plasticity and stabilizing post-stimulation oscillatory dynamics in neuronal populations.
June 2023	**Invited Speaker, CAIMS Annual Conference, University of New Brunswick, NB, Canada** Talk: Selective control of synaptic plasticity and stabilizing post-stimulation oscillatory dynamics in neuronal populations.
June 2022	**Speaker, Canadian Computational Neuroscience Spotlight, Canada**

Talk: Selective engagement of synaptic coupling using tACS.

Dec. 2016 **Speaker, 5th Basic and Clinical Neuroscience Congress, Tehran, Iran**

Talk: High-frequency networks determine effective connectivity in neural networks.

TEACHING AND MENTORING EXPERIENCE

2021-2025 **Multiple PhD and Master Students, University of Ottawa, and The Royal, Ottawa, ON, Canada.**

July-Aug. 2022 **FUSRP Project Supervisor, University of Ottawa, Ottawa, ON, Canada**

July 2020 **Teaching Assistant, Neuromatch Academy, Mashhad, Iran**

2013-2015 **Associate Supervisor, M.Sc. Thesis, University of Zanjan, Zanjan, Iran**

Student: M. Emamian.
Thesis: Honeycomb permanent magnetic lattice for ultra-cold atoms.

Fall 2014-2015 **Instructor, Quantum Mechanics, Abdor Rahman Sufi Razi Higher Education Institute, Zanjan, Iran.**

Fall 2014-2015 **Instructor, Basic Physics Laboratory, Institute for Advanced Studies in Basic Sciences, Zanjan, Iran.**

2012 **Workshop Mentor, Numerical Physics for Graduate Students of Physics, IASBS, Zanjan, Iran**

Software focus: MATLAB.

2012 **Instructor, Basic Physics and Laboratory, Industrial Management Institute, Mashhad, Iran**

2009-2010 **Teaching Assistant, Basic Physics I, University of Zanjan, Zanjan, Iran**

TECHNICAL SKILLS

Programming: Python, MATLAB, Fortran.

Software and toolboxes: EEGLAB, Brainstorm, SimNIBS, MATLAB App Designer, NESTApp, FSL.

Methods: computational modeling, dynamical systems, neural network modeling, TMS-EEG preprocessing and analysis, EEG/fMRI-informed modeling, effective connectivity, information-theoretic measures, machine learning, and statistical data analysis.

Research methods: non-invasive brain stimulation, tACS/tES/tCS, TMS, theta burst stimulation, neuroimaging-informed stimulation design, and subject-specific modeling.

AWARDS AND HONORS

2018 **Q1 Paper Publication Award, Cognitive Science and Technologies Council, Iran**

2016 **Travel Award for Ph.D. Students, OCNC2016, Jeju, South Korea**